

A Novel Interpretable Classification of Lumbar Spinal Stenosis Using a Cascade Deep Learning Approach and T2-Weighted MRI

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Inventory Category:	DL stenosis / multi-pathology
Comparability / Priority:	Very High Priority: Critical

Study Details & Methodology from Literature Inventory

Study Type / MRI View:	Interpretable cascade DL T2-weighted lumbar MRI
Dataset & Cohort:	None (Sample: None)
Disease / Target:	Lumbar spinal stenosis
Model / AI Method:	Three-stage cascade deep learning
Evaluation Metrics:	None
Key Finding / Relevance:	None
Thesis Context (Ch 2):	Very recent anatomy-aware and interpretable stenosis pipeline.

Abstract / Summary

OBJECTIVE Lumbar spinal stenosis (LSS) is a degenerative spinal condition characterized by the narrowing of the lumbar spinal canal, leading to back pain and disability. MRI remains the gold standard for LSS diagnosis, but diagnostic variability arises due to the lack of standardized imaging criteria. Recent advancements in artificial intelligence, particularly convolutional neural networks (CNNs), offer promising potential for automating LSS detection and classification. The aim of this study was to propose a novel 3-stage deep learning pipeline for automated LSS identification, classification, and grading using lumbar MRI, aiming to enhance diagnostic accuracy and consistency. **METHODS** Two datasets were used. The first dataset consisted of 17,440 MRI slices obtained in 640 patients (mean patient age 57.58 ± 12.47 years) and was used for model training. The second dataset consisted of 8000 slices and was used only as the external validation set. The proposed framework consists of 1) classification of images into sacral, lumbar, and thoracic regions; 2) region of interest detection; and 3) LSS grading (binary and multiclass). The 10-fold cross-validation method was used to avoid overfitting and improve generalization of the model. **RESULTS** The proposed model achieved an accuracy of 97.87% for binary classification and 95.52% for multiclass grading of LSS, outperforming state-of-the-art models. To validate the clinical relevance of the model's decision-making, gradient-weighted class activation mapping was used to visualize the key focus areas. **CONCLUSIONS** The proposed framework offers a reliable, interpretable, and effective tool for automated LSS detection and grading, with the potential for future improvements in underdiagnosis and multilevel spine disease analysis.